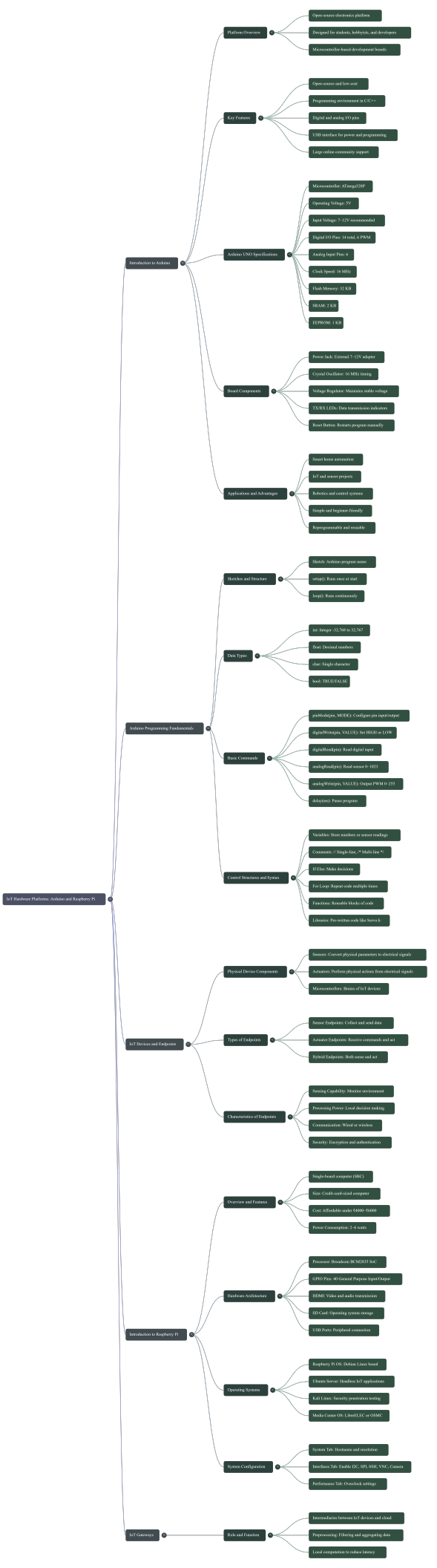

IoT Hardware Platforms: Arduino and Raspberry Pi
Introduction to Arduino
Platform Overview
Open-source electronics platform
Designed for students, hobbyists, and developers
Microcontroller-based development boards
Key Features
Open-source and low-cost
Programming environment in C/C++
Digital and analog I/O pins
USB interface for power and programming
Large online community support
Arduino UNO Specifications
Microcontroller: ATmega328P
Operating Voltage: 5V
Input Voltage: 7–12V recommended
Digital I/O Pins: 14 (of...
Analog Input Pins: 6
Clock Speed: 16 MHz
Flash Memory: 32 KB
SRAM: 2 KB
EEPROM: 1 KB
Board Components
Power Jack: External 7–12V adapter
Crystal Oscillator: 16 MHz timing
Voltage Regulator: Maintains stable voltage
TX/RX LEDs: Data transmission indicators
Reset Button: Restarts program manually
Applications and Advantages
Smart home automation
IoT and sensor projects
Robotics and control systems
Simple and beginner-friendly
Reprogrammable and reusable
Arduino Programming Fundamentals
Sketches and Structure
Sketch: Arduino program name
setup(): Runs once at start
loop(): Runs continuously
Data Types
int: Integer -32,768 to 32,767
float: Decimal numbers
char: Single character
bool: TRUE/FALSE
Basic Commands
pinMode(pin, MODE): Configure pin input/output
digitalWrite(pin, VALUE): Set HIGH or LOW
digitalRead(pin): Read digital input
analogRead(pin): Read sensor 0–1023
analogWrite(pin, VALUE): Output PWM 0–255
delay(ms): Pause program
Control Structures and Syntax
Variables: Store numbers or sensor readings
Comments: // Single line, /* Multi-line */
If-Else: Make decisions
For Loop: Repeat code multiple times
Functions: Reusable blocks of code
Libraries: Pre-written code like Servo.h
IoT Devices and Endpoints
Physical Device Components
Sensors: Convert physical parameters to electrical signals
Actuators: Perform physical actions from electrical signals
Microcontrollers: Brains of IoT devices
Types of Endpoints
Sensor Endpoints: Collect and send data
Actuator Endpoints: Receive commands and act
Hybrid Endpoints: Both sense and act
Characteristics of Endpoints
Sensing Capability: Monitor environment
Processing Power: Local decision making
Communication: Wired or wireless
Security: Encryption and authentication
Introduction to Raspberry Pi
Overview and Features
Single-board computer (SBC)
Size: Credit-card-sized computer
Cost: Affordable under ₹4000–₹6000
Power Consumption: 2–6 watts
Hardware Architecture
Processor: Broadcom BCM2835 SoC
GPIO Pins: 40 General Purpose Input/Output
HDMI: Video and audio transmission
SD Card: Operating system storage
USB Ports: Peripheral connection
Operating Systems
Raspberry Pi OS: Debian Linux based
Ubuntu Server: Headless IoT applications
Kali Linux: Security penetration testing
Media Center OS: LibreELEC or OSMC
System Configuration
System Tab: Hostname and resolution
Interfaces Tab: Enable I2C, SPI, SSH, VNC, Camera
Performance Tab: Overclock settings
IoT Gateways
Role and Function
Intermediaries between IoT devices and cloud
Preprocessing: Filtering and aggregating data
Local computation to reduce latency